

42.4 Regression splines and truncated power bases

A spline space is finite-dimensional, so fitting a spline is least squares with the columns of Eq. 42.3.1. Nothing new is needed to *do* it; what is needed is judgement about the two things the analyst now chooses, how many knots and where.

Definition 42.4.1 — Regression spline

Given a degree d and fixed interior knots $\kappa_1 < \dots < \kappa_K$ inside the range of the design, the **regression spline** model is the linear model with model matrix

$$\mathbf{X} = [\mathbf{1}, \mathbf{x}, \dots, \mathbf{x}^d, (\mathbf{x} - \kappa_1)_+^d, \dots, (\mathbf{x} - \kappa_K)_+^d],$$

of $d + 1 + K$ columns, and the **regression spline** is the least squares fit $\hat{f}(t) = \mathbf{x}(t)^\top \hat{\boldsymbol{\beta}}$. Other regressors, smooth terms and factors may be added as further columns.

Being an ordinary linear model, it has $n - (d + 1 + K)$ residual degrees of freedom at full rank; Theorem 12.1.1 gives the fitted curve an interval and Theorem 12.4.1 a band, both conditional on the knots; the leverages average $(d + 1 + K)/n$, which is what “ K knots cost K degrees of freedom” means; adding a knot is a one-degree-of-freedom nested test (Theorem 11.1.1) asking, by Exercise (B3), whether the d th derivative really jumps there; and choosing K by GCV (Definition 29.3.2) or AIC (Proposition 29.2.4) belongs to [Chapter 29](#).

42.4.1 What the knots control

Proposition 42.4.1 — Number, placement and basis

Fit a regression spline of degree d with interior knots $\kappa_1 < \dots < \kappa_K$ inside the range of the design, and write $\mathbf{M}_K$ for the projection onto the column space.

- (a) (*Number.*) If $\mathbf{X}$ has full rank, $\text{tr } \mathbf{M}_K = d + 1 + K$, so the leverages average $(d + 1 + K)/n$ and $\text{E}\|(\mathbf{I} - \mathbf{M}_K)\mathbf{Y}\|^2 = \sigma^2\{n - (d + 1 + K)\} + \|(\mathbf{I} - \mathbf{M}_K)\boldsymbol{\mu}\|^2$. Increasing K decreases both the squared bias and the residual degrees of freedom; the price is the total variance of the fitted curve, $\sigma^2 \text{tr } \mathbf{M}_K = \sigma^2(d + 1 + K)$, which rises by σ^2 per knot. Knots buy bias reduction at one degree of freedom each.
- (b) (*Rank.*) If each piece $[\kappa_k, \kappa_{k+1})$, $k = 0, \dots, K - 1$, and the last piece $[\kappa_K, b]$ contain at least $d + 1$ distinct design points, then $\mathbf{X}$ has full column rank. If some knot exceeds $\max_i x_i$, its column is identically zero and $\mathbf{X}$ does not.
- (c) (*Placement.*) For $d = 0$ the variance of the fit on interval k is σ^2/n_k , with n_k the number of design points there, so equalizing the counts minimizes the largest pointwise variance; quantile knots do that and equally spaced knots need not. For $d \geq 1$ the effect persists in the exact variance $\sigma^2 \mathbf{x}(t)^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}(t)$, which is large wherever an interval is sparsely occupied.
- (d) (*Basis.*) The columns of Eq. 42.3.1 are globally supported and increasingly parallel: beyond the last knot every one behaves like t^d , the angle between $(\mathbf{x} - \kappa_j)_+^d$

and $(\mathbf{x} - \kappa_k)_+^d$ tends to zero as $\kappa_j - \kappa_k \rightarrow 0$ (Exercise (C2)), and $\kappa(\mathbf{X})$ grows with K . A basis of the *same space* whose members are each supported on $d + 1$ consecutive intervals has a **basis condition number** — the worst ratio between the size of a coefficient vector and the size of the function it builds — bounded by a constant depending only on d , whatever the knots (de Boor, 2001). For a design with points in every basis function's support, $\kappa(\mathbf{X})$ then does not grow with K ; without one nothing is claimed, and $\mathbf{X}$ can be singular.

Proof.

- (a) $\mathbf{M}_K$ is an orthogonal projection, so its trace is its rank (Proposition 6.3.4), namely $d + 1 + K$ by Theorem 42.3.1 at full rank; the expected residual sum of squares is the decomposition of Section 9.1. Adding a knot enlarges the column space, so the squared bias cannot increase while $\text{tr}(\mathbf{I} - \mathbf{M})$ falls by one.
- (b) Suppose $\mathbf{X}\mathbf{c} = \mathbf{0}$. By Theorem 42.3.1 the function $g = \sum_j c_j \phi_j$ built from the same coefficients is a spline vanishing at every design point. On each piece g is a polynomial of degree at most d with $d + 1$ distinct roots there, hence identically zero there; so $g \equiv 0$ and, the basis being independent, $\mathbf{c} = \mathbf{0}$. For the second statement, $(x_i - \kappa)_+^d = 0$ for every i when $\kappa > \max_i x_i$.
- (c) For $d = 0$ the columns are the orthogonal indicators of the intervals, so $\mathbf{X}^\top \mathbf{X} = \text{diag}(n_0, \dots, n_K)$ and the fit on interval k is $\bar{y}_k$, with variance σ^2/n_k . Over counts summing to n , the largest σ^2/n_k is smallest when they are as equal as possible, which quantile knots achieve up to rounding. For $d \geq 1$ the variance formula is Theorem 12.1.1, and that it is large over a sparse interval is not an identity but an observation.
- (d) The first statement is Exercise (C2). The bound on the basis condition number is quoted from de Boor (2001) and not proved here; that it controls $\kappa(\mathbf{X})$ needs the design to occupy every support, which is the next paragraph.

■

Part (c) is demonstrated in [Example \(A skewed design\)](#) and part (d) in [Example \(Truncated powers against a local basis\)](#), whose nearly uniform design meets the proviso; the local basis is built in [Chapter 43](#).

Part (b) has a sharp form. In the local basis the basis matrix has full rank *if and only if* each basis function's support contains a design point — the Schoenberg–Whitney condition (Schoenberg and Whitney 1953), which is also the proviso in (d). Either way the rule is: no knot where there are no data, and no two knots with a nearly empty interval between them.

42.4.2 Six bases for one trend

Example. (Trend bases for the carbon dioxide series)

Return to the $n = 521$ monthly means with the four seasonal columns fixed, put $K = 6$ interior knots at the quantiles $k/7$ of the time variable, and compare six trend bases.

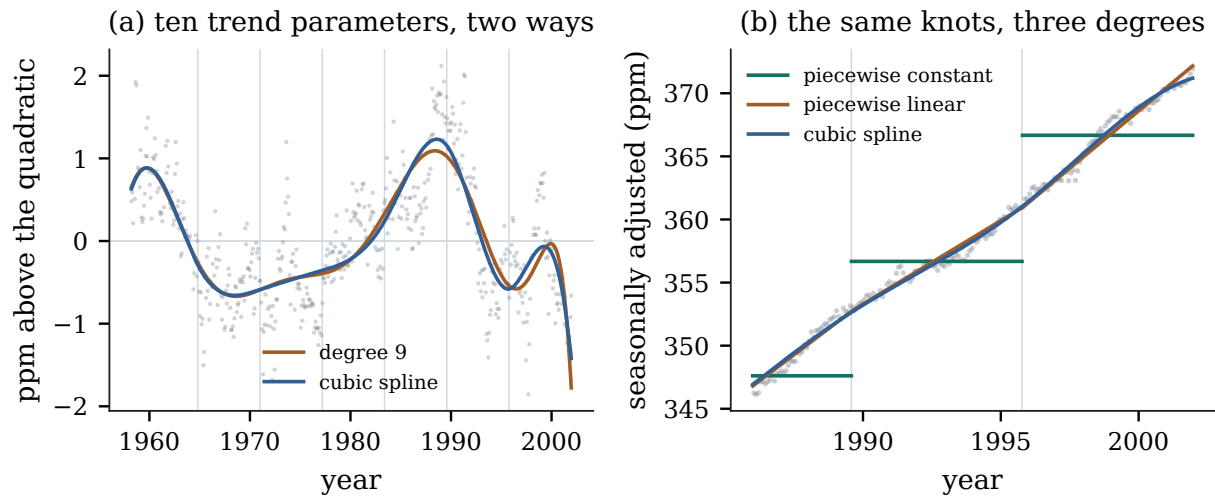


Figure 42.4.1: (a) The degree-9 polynomial trend and the cubic spline with six quantile knots, both with ten trend parameters, as departures from the quadratic trend. (b) Three fits on the same knots (vertical lines) over the last fifteen years, seasonally adjusted.

Trend basis	p	Residual SS	$\hat{\sigma}$	GCV	$\max h_{ii}$
quadratic	7	297.03	0.7602	0.5858	0.0256
degree 9	14	115.57	0.4774	0.2343	0.2054
degree 16	21	92.44	0.4300	0.1926	0.4546
piecewise constant	11	3081.18	2.4580	6.1718	0.0215
piecewise linear	12	116.63	0.4787	0.2345	0.0532
cubic spline	14	108.48	0.4626	0.2199	0.1494

Read the middle rows against each other. At the same $p = 14$ the cubic spline fits better than the degree-9 polynomial (108.48 against 115.57) and spreads its influence more evenly (largest leverage 0.1494 against 0.2054); panel (a) of Figure 42.4.1 shows the two parting where the polynomial has to bend. The step function is a disaster; the broken line is almost as good as the cubic spline with two fewer parameters, at the cost of visible kinks (panel (b)).

In-sample, GCV prefers the degree-16 polynomial to everything else (0.1926) — honest about prediction at the design points and misleading about the fit, since that model has $\max h_{ii} = 0.4546$ and $\kappa(\mathbf{X}) = 5.39 \times 10^5$. Among the piecewise bases the cubic spline is the worst conditioned: $\kappa(\mathbf{X}) = 7.72 \times 10^3$ against 9.75×10^1 .

```
import numpy as np
import statsmodels.api as sm

co2 = sm.datasets.co2.load_pandas().data["co2"]
monthly = co2.resample("MS").mean().dropna()
y = monthly.to_numpy()
t = (monthly.index.year + (monthly.index.month - 0.5) / 12).to_numpy()
n = len(y)
lo, hi = t.min(), t.max()
u = lambda s: 2 * (s - lo) / (hi - lo) - 1
```

```

K = 6
knots = np.quantile(t, np.arange(1, K + 1) / (K + 1))      # quantile knots
print("knots at", np.round(knots, 2))

def polynomial(s, d):
    return np.vander(u(s), d + 1, increasing=True)

def step(s):
    0
    piece = np.digitize(s, knots)
    return np.column_stack([(piece == k).astype(float) for k in range(K + 1)])

def broken_line(s):
    linear
    return np.column_stack([np.ones_like(s), u(s)]
                           + [np.clip(u(s) - u(k), 0.0, None) for k in knots])

def cubic_spline(s):
    d = 3
    return np.column_stack([u(s) ** j for j in range(4)]
                           + [np.clip(u(s) - u(k), 0.0, None) ** 3 for k in knots])

def with_season(f):
    return lambda s: np.column_stack([f(s), np.cos(2 * np.pi * s), np.sin(2 * np.pi
    * s),
                                     np.cos(4 * np.pi * s), np.sin(4 * np.pi * s)])

bases = {"quadratic": lambda s: polynomial(s, 2), "degree 9": lambda s: polynomial(s
, 9),
        "degree 16": lambda s: polynomial(s, 16), "piecewise constant": step,
        "piecewise linear": broken_line, "cubic spline": cubic_spline}

```

[Run this code online](#)

```

for name, f in bases.items():
    X = with_season(f)(t)
    p = np.linalg.matrix_rank(X)
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    resid = y - X @ beta
    rss = resid @ resid
    lev = np.einsum("ij,ij->i", X @ np.linalg.pinv(X.T @ X), X)
    print(f"{name:20s} p = {p:2d}    RSS = {rss:8.2f}    sigma = {np.sqrt(rss / (n - p
    )):.4f}")
    f"    GCV = {n * rss / (n - p) ** 2:.4f}    max leverage {lev.max():.4f}"
    f"    cond = {np.linalg.cond(X):.1e}"

```

[Run this code online](#)

Locality shows up in the same fits: moving the December 2001 value by 10 ppm moves the March 1958 trend by 0.197 ppm under the degree-12 polynomial but 0.004 under the cubic spline and 0.001 under the step function, where only the shared seasonal columns keep it from being zero.

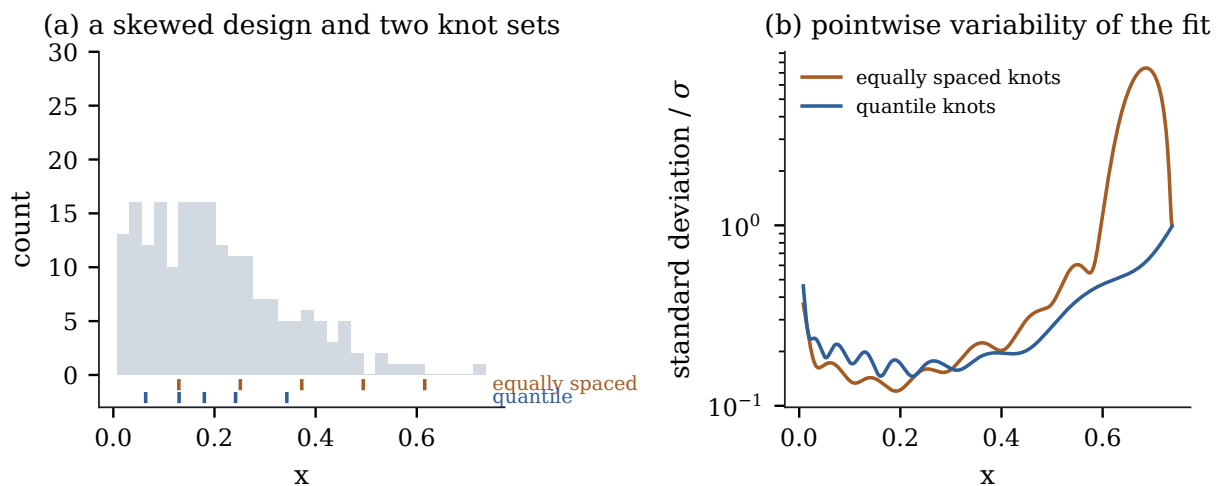


Figure 42.4.2: (a) A skewed design of 200 points with five equally spaced and five quantile knots. (b) Pointwise standard deviation of the fit, in units of σ , log scale.

42.4.3 Where to put them

Example. (A skewed design)

Take $n = 200$ design points drawn once, with a fixed seed, from a skewed distribution on $[0, 1]$, and fit a cubic spline with $K = 5$ interior knots. Equally spaced knots leave 67, 71, 35, 21, 5, 1 points in the six intervals; quantile knots leave 34, 33, 33, 33, 33, 34. Both have full rank — the truncated powers are globally supported, so a nearly empty interval does not kill a column — but it does kill precision. The exact pointwise standard deviation of the fit, in units of σ , reaches 7.452 with the equally spaced knots against 0.987 with the quantile knots; at the median of the design the quantile knots are slightly *worse* (0.173 against 0.123), having spent four of five knots there.

Quantile knots are a defensible default, not a universal optimum: they make the worst case tolerable. Where the regression function is known to vary fast in a sparse region the knots should follow the function — a choice made from outside the data, or paid for as a selection (Proposition 29.5.1).

```
import numpy as np

rng = np.random.default_rng(42)
n = 200
x = np.sort(rng.beta(1.5, 6.0, n))           # a skewed design on [0, 1]
K, d = 5, 3

equal = np.linspace(x.min(), x.max(), K + 2)[1:-1]   # equally spaced knots
quantile = np.quantile(x, np.arange(1, K + 1) / (K + 1))

def spline_design(s, knots):
    """Truncated power basis for degree d with the given interior knots."""
    return np.column_stack([s ** j for j in range(d + 1)]
                           + [np.clip(s - k, 0.0, None) ** d for k in knots])
```

```
def counts(knots):
    edges = np.r_[-np.inf, knots, np.inf]
    return [int(((x >= a) & (x < b)).sum()) for a, b in zip(edges[:-1], edges[1:])]

print("equally spaced knots:", np.round(equal, 3), " points per interval", counts(
    equal))
print("quantile knots:      ", np.round(quantile, 3), " points per interval",
    counts(quantile))
```

[Run this code online](#)

```
grid = np.linspace(x.min(), x.max(), 500)
for name, knots in (("equally spaced", equal), ("quantile", quantile)):
    X = spline_design(x, knots)
    G = np.linalg.inv(X.T @ X)
    Xg = spline_design(grid, knots)
    sd = np.sqrt(np.einsum("ij,jk,ik->i", Xg, G, Xg)) # in units of sigma
    print(f"{name:15s} knots: standard deviation of the fit at the median {np.interp(
        np.median(x), grid, sd):.3f},"
        f" largest {sd.max():.3f} at x = {grid[sd.argmax():.3f]}")
```

[Run this code online](#)

Idea. Rules of thumb that survive scrutiny, for a fixed-knot regression spline:

- use a low degree, almost always cubic, and control flexibility with K ;
- place knots at quantiles unless something outside the data says otherwise, keeping several observations between consecutive knots;
- compare a few values of K by GCV or AIC rather than searching finely;
- report the fit as a curve with a band, never as coefficients, which depend on the basis and mean nothing alone.

The middle two leave a discrete, data-dependent choice. [Chapter 43](#) removes it by taking K large and adding a penalty, making flexibility continuous (Definition 43.3.1).

42.4.4 Why this basis is not the one to compute with

Example. (Truncated powers against a local basis)

For the carbon dioxide design rescaled to $[-1, 1]$, take cubic splines with K quantile knots and compare the condition number of the truncated power basis with that of the local (B-spline) basis for the *same* space, verified by checking that the two sets of columns together have rank $K + 4$.

K	4	8	16	32
truncated powers	1.75×10^3	2.56×10^4	3.65×10^5	4.97×10^6
local basis	5.30	5.18	5.04	5.00

The truncated power condition number multiplies by about fifteen for every doubling of K ; the local basis does not move. By $K = 32$ the normal equations (Theorem 6.4.2) in the truncated power basis have lost about thirteen digits (Eq. 10.1.2), the same fit in the local basis one. This design is nearly uniform, so every basis function's support is well occupied — the proviso of part (d).

The remedy is that of [Section 42.2](#): the space is fine and the basis is bad. The local basis is built in [Chapter 43](#) (Definition 43.2.1 and Proposition 43.2.1). The truncated power basis keeps its place in the theory — Theorem 42.3.1 is easiest to prove with it, and θ_k means a jump in the d th derivative — but it should not be what a program forms.

42.4.5 Honest limits of a fitted curve

Three warnings close the chapter, none of them about arithmetic. *Outside the range there is nothing*: a cubic spline continues as a cubic beyond its last knot, and refitting to the months up to 1990 and predicting 1991–2001 gives root mean squared errors of 2.71 ppm for the broken line, 3.99 for the quadratic, 46.11 for the cubic spline and 923.15 for the degree-9 polynomial — almost the reverse of the ranking inside the data.

A curve is a conditional mean, not a mechanism: dropping linearity removes a functional-form assumption and nothing else, and Theorem 25.4.1 is still what licenses a causal reading. And *a flexible fit is no substitute for a good parameterization*: where the relationship is known to be a power law, the transformations of [Chapter 22](#) give a two-parameter model that a ten-parameter spline can only approximate and that extrapolates on a principle (Theorem 22.3.2). A spline is often best used diagnostically: fit it, look at the shape, then ask whether a parametric form with meaning would do.

42.4.6 Exercises

42.4.6.1 A. Check your understanding

Exercise A1.

A cubic regression spline with K quantile knots is fitted to $n = 200$ observations together with three other regressors and an intercept. Give the residual degrees of freedom and the average leverage as functions of K , and the K at which the fit interpolates.

Exercise A2.

Three analysts fit a cubic spline with the same six knots, in the truncated power basis, in the local basis, and as free cubics under the continuity constraints. Which outputs agree exactly, which to rounding, and which disagree?

42.4.6.2 B. Practice

Exercise B1.

Fit cubic splines to the carbon dioxide series with $K = 2, 4, \dots, 20$ quantile knots and plot GCV against K , then the leave-one-out criterion computed by the leverage shortcut of Theorem 29.3.2. Where do the two disagree, and why should you not report the F test comparing the selected K with $K = 2$?

Solution. The two have the same minimizer here or nearly so; GCV replaces each $1 - h_{ii}$ by its average $1 - p/n$, which matters when the leverages are uneven, as at the boundary. The F test is invalid because K minimized a criterion computed from the same data: the residual mean square is biased downwards and the statistic is not F (Proposition 29.5.1).

Exercise B2.

Compute the pointwise 95% band for the cubic spline fit of [Example \(Trend bases for the carbon dioxide series\)](#) and the simultaneous Working–Hotelling band (Theorem 12.4.1). By what factor is the simultaneous band wider, and what does each band cover?

Solution. The ratio of half-widths is $\sqrt{pF_\alpha(p, n-p)}/t_{\alpha/2}(n-p)$, about two and a half for $p = 14$ and $n = 521$. The pointwise band covers $E(Y | x = t)$ at a single pre-chosen t with probability 0.95, the simultaneous band the whole curve. Neither covers anything if the true mean is not in the spline space: both are conditional on the model, and the model includes the knots.

Exercise B3.

A model has a cubic spline in x and a linear term in z . Show that the coefficient of z is the slope of the regression of $\mathbf{y}$ on the part of $\mathbf{z}$ orthogonal to the spline space, and say what it would mean for that part to be nearly zero.

Solution. This is Theorem 6.6.2 with $\mathbf{X}_2$ the spline columns. If $\mathbf{z}$ is nearly a spline function of x — because z is a smooth transformation of it, or because K is so large that the spline space reproduces almost anything — then $(\mathbf{I} - \mathbf{M}_2)\mathbf{z}$ is short, the coefficient's variance huge, and the two terms not separately identified: the additive-model version of collinearity, taken up in [Chapter 44](#) (Proposition 44.2.3).

42.4.6.3 C. Going deeper

Exercise C1.

For a regression spline the fitted value at t is $\sum_i \ell_i(t)y_i$ with $\ell_i(t) = \mathbf{x}(t)^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i$. Show that $\sum_i \ell_i(t) = 1$ and $\sum_i \ell_i(t)(x_i - t)^r = 0$ for $r = 1, \dots, d$ at every t , because $\mathcal{S}_d(\kappa)$ contains $\mathcal{P}_d$ and a projection reproduces its own space. Deduce that the fit reproduces polynomials of degree d exactly, and interpret $\sum_i \ell_i(t)^2$.

Exercise C2.

Suppose K minimizes GCV over $K \in \{1, \dots, 20\}$ and the usual pointwise band is then reported for the selected fit. Simulate 2000 data sets from a known smooth f with normal errors and estimate the coverage of the nominal 95% band at the quartiles of

the design, explaining the pattern in terms of Proposition 29.5.1 and of the bias a selected K leaves behind.